

Rishi Patel

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EDUCATION

Toronto Metropolitan University

September 2022 - May 2027

Bachelor of Engineering in Computer Engineering

- Relevant Courses: DSA, OOP, Software Systems, Electronic Circuits, Physics, Discrete Math, Digital Systems

EXPERIENCE

Clubby - Software Engineer/Co-Founder

May 2024 – Present

- Built a platform using **Flutter**, **Firestore** and **Database** to centralize university club and event management.
- Leading a team of 5 engineers using **Scrum/Agile** to accelerate feature deployment and development by 20%.
- Implemented **user-friendly** features to help students discover and manage university clubs and events, facilitate in-club and event communication, and streamline marketing efforts, boosting engagement and productivity by estimated 30%.

LaunchX Dev - Product Design & Development Intern

September 2023 – December 2023

- Led design validation for the Yume app startup, conducting 5+ rounds of **manual** and **automated** testing that improved usability and reduced user issues by 40%, achieved from collaborating with **cross-functional** teams through Asana.
- Optimized GoDaddy website routing for 3 sites, increasing traffic by 15% and improving navigation.
- Collaborated with a full-stack engineer through 2 months of job shadowing, gaining hands-on experience in **full app development lifecycle** from front-end (**React.js/CSS**) to back-end (**Node.js/SQL**), streamlining product deployment.

RP Game Services - Software Engineer - Contracts/Commissions

May 2022 – January 2024

- Developed comprehensive server plugins for gaming platforms PocketMine (**PHP**) and Bukkit (**Java**), featuring in-game functionalities, purchasable website integrations, and interactive Discord linkage, used by 500+ players.
- Established **server-client/server-server** communication (97% accuracy) and a **robust MySQL** DBMS, reducing **SQL injection risks** by 95% with prepared statements and boosting overall performance with **asynchronous** operations.
- Provided **comprehensive** documentation and end-user support for program development, plugin **API** and libraries.

Skills

Languages: Python, Java, C, C++, MySQL, MongoDB, SQLite, Assembly, Verilog, Javascript, Dart, HTML, CSS

Tools: Git, Bash, JetBrains IDEs, VSCode, Powershell, Quartus, ModelSIM, MultiSIM, MATLAB, Firebase, SCRUM, SOLID

Hardware/OS: Arduino, ESP32, Oscilloscope, Function Generator, Circuit Design, Windows, Linux (Ubuntu)

Testing/Frameworks: Flutter, Keras, Tensorflow, Jupyter, Pandas, Pytest, JUnit, Docker, React.js, Node.js, Next.js, Express.js

PROJECTS

SmartLamp - Arduino, C, HTML, CSS, ESP32

[LINK](#)

- Designed a smart lamp system controlled via either **sound sensor** or buttons on a **WiFi-hosted** server
- Developed a web-based interface using **HTML/CSS** for remote lamp control on WiFi and integrated sound detection for **hands-free operation**. Obtained 90% response accuracy communicating with the **ESP32** and lamp.

UniManagement - Java, JSwing, MongoDB, JUnit

[LINK](#)

- Developed a university management system in **Java** for students, professors, and TAs with a seamless **JSwing UI**.
- Incorporated **MongoDB** for efficient storage of authentication and role-based data, tested for over 25k users utilizing **JUnit**, **testcontainers** and **mockito** for **CRUD** integration tests and service logic coverage.
- Enhanced quality, scalability, and maintainability by 25% following **SOLID** principles and design patterns.

LiteScript - C++

[LINK](#)

- Designed and implemented LiteScript, a programming language featuring a custom syntax, leveraging C++ to create a lexer and parser that efficiently translates LS source code into an abstract syntax tree (AST) for analysis and execution.
- Engineered a robust compiler architecture, allowing for seamless code compilation on Assembly or interpretation through commands, enhancing usability and developer experience.

Stock Prediction - Python, Keras, Pandas, Jupyter, SQLite

[LINK](#)

- Developed a **Python** application to predict stock market and cryptocurrency trends with 60% accuracy, utilizing a machine learning model with a **5-layer neural network**, incorporating **LSTMs** in **Keras** and **Scikit-learn**.
- Trained the model on data from 100+ stocks and cryptocurrencies scraped via **yfinance**, preprocessed for accuracy.
- Integrated **SQLite3** for user authentication and stock favoriting, deploying the application seamlessly on **Streamlit**.

R-GameHub - Node.js, React.js, CSS, Express.js, MySQL, Tensorflow, OpenAI

- Developed a multi-game platform incorporating single-player and real-time multiplayer modes with queuing using **Node.js**, **React.js** and **SocketIO**, achieving sub-300ms server-client communication using **Express** and **Axios (REST)**
- Leveraged **TensorFlow** Handpose for camera-based finger detection and **OpenAI** for calculating optimal moves